

“We are currently working on a future for iText that goes beyond PDF”

Open source has not just helped some established organisations to grow further but also empowered many startups to compete against big enterprises. iText is one such example.

The Belgium-based company brought out the world's most used and actively developed PDF library by deploying open source technologies. **Jagmeet Singh** of *OSFY* talked to *iText founder Bruno Lowagie* to get some insights about his venture and his views on the growth of open source. Edited excerpts from the interview:

Q How did you get the idea to launch iText?

I got my first computer at the age of 12. That was 34 years ago (in 1982) and things were very different back then. Today, people get a computer and they can start working right away. In those days, there wasn't that much software around and what was available was very expensive. At the age of 14, I had already written my own database system and my own text editor. I didn't know about the Internet back then; I was the only kid who owned a computer in my town.

At the age of 18, I went to college, but I didn't study computer science or software engineering. I picked civil architectural engineering. I learned how to design buildings, not software. I returned to software development only after I graduated. I changed jobs four times in the first three years of my career — working for different software integrators first, then finally for Ghent University. During my spare time, I continued working on personal projects, but none of them were successful until I released iText.

My job at Ghent University consisted of rewriting the software for the student administration. Previously, professors had to enter the grades of the students on a diskette using a Clipper application that only worked on DOS and that could be used to print lists on an HP printer (and only on an HP printer). These diskettes were then passed on to an administrator who extracted the data and uploaded it to a central database. I promised Ghent University that I would offer them an intranet application that could be used on any computer using nothing but a browser. I also promised that I would provide lists in the PDF format so that they wouldn't depend on one particular brand of printers.

The year was 1998. I used Java servlets to create the intranet application. Java has existed since 1995 and the first PDF specification was released in 1993. I was convinced that I would be able to find PDF software that could be used

to create PDF files on-the-fly in a server application. I was wrong; in those days, PDF documents were created on the desktop by human beings who had access to a GUI. I didn't find any Java PDF software that met the requirements of my project — I needed to create PDFs on the server, without a GUI and without human intervention. Moreover, the software had to be fast; when people asked for a list with 20,000 student records, they expected that list to be available in fractions of a second. Nobody wants to wait for minutes to get a document from the Internet.

So I started reading the PDF specification during the Christmas holidays of 1998 and I began to hate PDF. Reading the PDF specification to understand PDF is like reading a dictionary in the hope you'll learn a language. I succeeded in writing the first PDF library in six weeks' time. It served its purpose, but a developer really needed to understand PDF inside-out if he or she wanted to use that first library. That wasn't trivial; I was the only person on my project team who understood the code. As a result, I was the only person who could debug and maintain that part of the project. That's not a healthy situation.

After having worked with that library for about a year, I suddenly saw the light — I started to understand and love PDF. I realised that my first library was all wrong and I wanted to rewrite it from scratch. My employer didn't understand why I wanted to do that — why fix something that isn't broken? But I persisted and I started writing a new library in my spare time. I called this new library iText, and I released the first version as open source software in February 2000.

Q What distinguishes iText as an open source library in the market?

The strength of iText is its ease of use. Thanks to iText, developers could start creating PDFs without having to know